# STAT A411/611 Lecture Notes

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**Office hours:** MWF 12:00–12:45 **Class:** 2:30–3:45

## Lecture Plan and Review

**Today:** Sections 3.1 and 3.2.

**Review:** Monday's Chapter 2 material, Sections 2.1–2.3.

**Reading scope:** Skip Section 2.4.

### Section 2.1: Axioms and Addition Rules

$$\text{(A1)} \quad P(A) \geq 0, \qquad \text{(A2)} \quad P(S) = 1.$$

For mutually exclusive events, Axiom 3 is the **Special Addition Rule (SAR)**:

$$\text{(A3)} \quad P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i).$$

**Complement Rule (CR):** $P(A) = 1 - P(A^c)$.

For general events, the **General Addition Rule (GAR)** is

If $A_1, A_2, \ldots, A_n$ are any events, then

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{i=1}^{n} P(A_i) - \sum_{i<j} \sum P(A_i \cap A_j) + \sum_{i<j<k} \sum \sum P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P(A_1 \cap \cdots \cap A_n).$$

Indices range from 1 to $n$; each intersection is counted once. For mutually exclusive events, these terms vanish.

### Section 2.2: Conditional Probabilities

**Examples.**

$P(B \mid A)$ is the probability of $B$ given that $A$ has occurred. The **Conditional Probability Rule (CPR)** is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad \big(P(A) > 0\big).$$

### Section 2.3: General Multiplication Rule (GMR)

If $A_1, A_2, \ldots, A_n$ are any events, then

$$P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1 \cap A_2) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Under independence, the conditional probabilities become the corresponding unconditional probabilities. This gives the **Special Multiplication Rule (SMR)**:

$$P\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} P(A_i).$$

## 3 Chapter 3 — Discrete Random Variables and Their Probability Distributions

We now cover Sections 3.1–3.8, beginning with Sections 3.1 and 3.2.

### 3.1 Section 3.1 — Random Variables

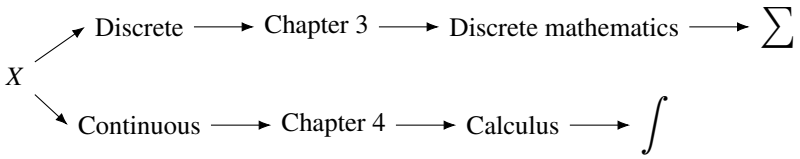
An experiment is performed and an outcome is observed.

**Definition.** A **random variable** is a rule that assigns a number to every possible outcome of an experiment.

$$X = \text{the random variable}, \qquad \mathcal{X} = \text{the set of possible values of } X.$$

A random variable is **discrete** if its set of possible values can be listed, that is, if $\mathcal{X}$ is a **countable set**.

$$\{1, 2, 3\}, \quad \{0, 1, 2, 3, \ldots\} \quad \text{discrete examples,}$$
$$[0, 1], \quad (0, \infty) \quad \text{continuous examples.}$$




**Example: Three Tosses of a Balanced Coin**

Toss a balanced coin three times and let $X$ be the number of heads. The possible values are

$$\mathcal{X} = \{0, 1, 2, 3\}.$$

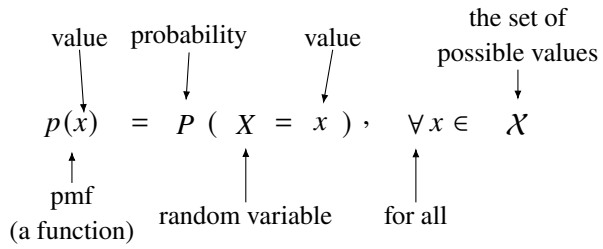
Then $X$ is a discrete random variable.

### 3.2 Section 3.2 — Probability Distributions

**Definition.** The **probability mass function (pmf)** of a discrete random variable $X$ is

$$\boxed{p(x) = P(X = x), \qquad \forall\, x \in \mathcal{X}.}$$

The notation distinguishes the function, the probability, the random variable, and its value:




A valid pmf satisfies

$$\text{(i)} \quad p(x) \geq 0 \qquad \forall x \in \mathcal{X}, \qquad \text{(ii)} \quad \sum_{x \in \mathcal{X}} p(x) = 1.$$

The pmf describes the probability distribution of $X$.

**Example: The Larger of Two Dice**

**Find the pmf of $X$.**

Roll a fair six-sided red die and a fair six-sided blue die. Let $X$ be the **largest** of the two upward-facing numbers. For example,

$$R = \boxed{\cdot}, \qquad B = \boxed{\cdot\,\cdot}\,, \qquad \Longrightarrow \qquad X = 5.$$

The possible values are $\mathcal{X} = \{1, 2, 3, 4, 5, 6\}$. The 36 equally likely ordered outcomes give the following maximum values:

| $B \backslash R$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | 1 | (2) | 3 | 4 | 5 | 6 |
| 2 | (2) | (2) | 3 | 4 | 5 | 6 |
| 3 | 3 | 3 | 3 | 4 | 5 | 6 |
| 4 | 4 | 4 | 4 | 4 | 5 | 6 |
| 5 | 5 | (5) | 5 | 5 | 5 | 6 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 |

The three circled entries give $X = 2$; the dashed entry shows the example $R = 2$, $B = 5$.

Using the Classical Formula,

$$p(1) = P(X = 1) = \frac{1}{36}, \qquad p(2) = P(X = 2) = \frac{3}{36}.$$

Continuing the same count gives

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $p(x)$ | 1/36 | 3/36 | 5/36 | 7/36 | 9/36 | 11/36 |

Thus

$$\boxed{p(x) = \frac{2x - 1}{36}, \qquad x = 1, 2, 3, 4, 5, 6,}$$

and

$$\sum_{x=1}^{6} p(x) = \frac{1+3+5+7+9+11}{36} = 1.$$

**Cumulative Distribution Function (CDF)**

The **cumulative distribution function** is

$$\boxed{F(x) = P(X \le x), \qquad -\infty < x < \infty.}$$

Its properties include $F(x) \ge 0$ for every real $x$, and it is **nondecreasing**:

$$a \le b \qquad \Longrightarrow \qquad F(a) \le F(b).$$

**Example: CDF of a Fair Six-Sided Die**

Roll one fair six-sided die and let $X$ be the upward-facing number. The CDF is defined for all real $x$, not only for the possible die values. **Find and graph the CDF of $X$.**

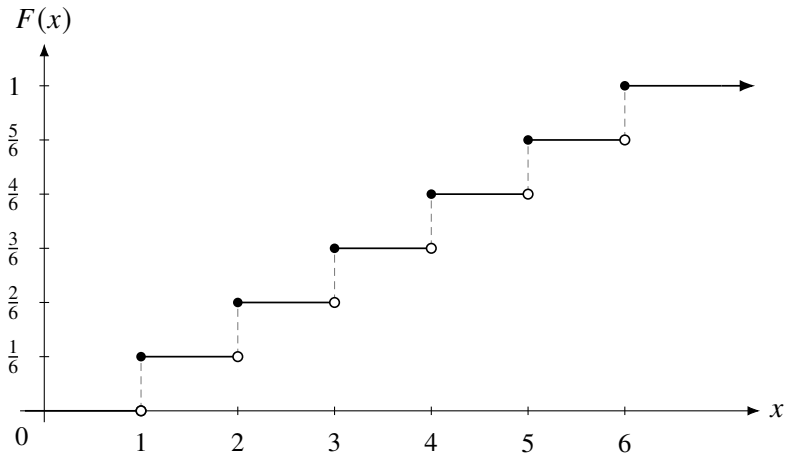
$$F(x) = P(X \le x) = \begin{cases} 0, & x < 1, \\ \frac{1}{6}, & 1 \le x < 2, \\ \frac{2}{6}, & 2 \le x < 3, \\ \frac{3}{6}, & 3 \le x < 4, \\ \frac{4}{6}, & 4 \le x < 5, \\ \frac{5}{6}, & 5 \le x < 6, \\ 1, & x \ge 6. \end{cases}$$

For example,

$$F(0) = P(X \le 0) = 0,$$

$$F(1) = P(X \le 1) = P(X = 1) = \frac{1}{6},$$

$$F(1.99) = P(X \le 1.99) = P(X \le 1) = \frac{1}{6}.$$



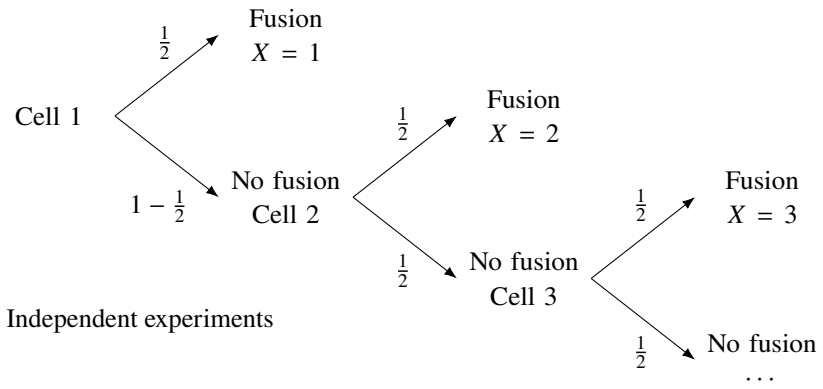

**a STEP FUNCTION**

The CDF stays constant between consecutive integers. Each new step includes its left endpoint and excludes its right endpoint.

**Homework:** Problems 2, 4, and 6, p. 81.

**Example 3.2.1: Cells Exposed Until Fusion**

Let $X$ be the number of cells exposed until a fusion takes place. Each cell produces fusion with probability $1/2$, and the experiments are independent.




**a)** Find the pmf of $X$.

**b)** Show that the function in (a) is a valid pmf.

**c)** Find the CDF of $X$.

**d)** Find $P(X \geq 10)$.

**e)** Find $P(5 \leq X \leq 25)$.

**(a) Find the Probability Mass Function**

The possible values are $\mathcal{X} = \{1, 2, 3, \ldots\}$. Write $C_i = F$ for fusion at cell $i$, and $C_i \neq F$ for no fusion at that cell.

$$p(1) = P(X = 1) = P(C_1 = F) = \frac{1}{2}.$$

For fusion at the second cell, the first cell must not fuse:

$$\text{IND.} \longrightarrow \text{SMR}$$

$$\begin{aligned} p(2) = P(X = 2) &= P(C_1 \neq F \ \cap \ C_2 = F) \\ &\overset{\text{SMR}}{=} P(C_1 \neq F)P(C_2 = F) \\ &\overset{\text{CR}}{=} \big(1 - P(C_1 = F)\big)P(C_2 = F) \\ &= \left(1 - \frac{1}{2}\right)\frac{1}{2} = \boxed{\frac{1}{4}}. \end{aligned}$$

Similarly,

$$\begin{aligned} p(3) = P(X = 3) &= P(C_1 \neq F \ \cap \ C_2 \neq F \ \cap \ C_3 = F) \\ &\overset{\text{SMR}}{=} \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{2}\right)\frac{1}{2} = \boxed{\frac{1}{8}}. \end{aligned}$$

**The General PMF**

Fusion first occurs at cell $x$ precisely when the first $x-1$ cells do not fuse and cell $x$ does:

$$\text{IND.} \longrightarrow \text{SMR}$$

$$\begin{aligned} p(x) = P(X = x) &= P(C_1 \neq F \ \cap \ C_2 \neq F \ \cap \ \cdots \ \cap \ C_{x-1} \neq F \ \cap \ C_x = F) \\ &\overset{\text{SMR}}{=} P(C_1 \neq F)P(C_2 \neq F)\cdots P(C_{x-1} \neq F)P(C_x = F) \\ &= \underbrace{\left(1-\frac{1}{2}\right)\left(1-\frac{1}{2}\right)\cdots\left(1-\frac{1}{2}\right)}_{x-1 \text{ times}}\frac{1}{2} \\ &= \left(\frac{1}{2}\right)^{x-1}\frac{1}{2} = \boxed{\left(\frac{1}{2}\right)^{x}}, \qquad x = 1, 2, 3, \ldots. \end{aligned}$$

**(b) Verify That This Is a Valid PMF**

**I. Nonnegativity.** $\forall\, x \in \mathcal{X}$,

$$p(x) = \left(\frac{1}{2}\right)^{x} \geq 0.$$

**II. Total probability.** We must show that

$$\sum_{x=1}^{\infty} p(x) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1.$$

The first partial sums are

$$\underbrace{\frac{1}{2}}_{0.5}, \qquad \underbrace{\frac{1}{2}+\frac{1}{4}}_{0.75}, \qquad \underbrace{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}}_{0.875}, \qquad \ldots$$

Does the infinite sum equal 1, 2, or diverge to $\infty$? A geometric-series calculation settles the question.

**Side Note: Geometric Sequences and Series**

For a real number $a$, the geometric sequence is

$$a^0,\ a^1,\ a^2,\ a^3, \ldots.$$

The finite geometric sum is

$$\boxed{\sum_{k=0}^{n} a^k = \frac{a^{n+1}-1}{a-1}} \qquad (a \neq 1).$$

For nonnegative $a$, the infinite sum satisfies

$$\sum_{k=0}^{\infty} a^k = \begin{cases} \dfrac{1}{1-a}, & 0 \le a < 1, \\ \infty, & a \ge 1. \end{cases}$$

**Proof of the Finite Geometric-Sum Formula**

Let

$$S = \sum_{k=0}^{n} a^k = 1 + a + a^2 + a^3 + \cdots + a^n.$$

Multiply by $a$ and subtract the original sum:

$$\begin{array}{rl} aS = & a + a^2 + a^3 + \cdots + a^n + a^{n+1} \\ S = 1 + & a + a^2 + a^3 + \cdots + a^n \\ \hline aS - S = -1 & \qquad\qquad\qquad\qquad + a^{n+1} \end{array}$$

All the middle terms cancel. Consequently,

$$\begin{aligned} aS - S &= a^{n+1} - 1, \\ (a-1)S &= a^{n+1} - 1, \\ S &= \boxed{\frac{a^{n+1}-1}{a-1}}, \qquad a \ne 1. \end{aligned}$$

**Proof of the Infinite Geometric-Series Formula**

For $0 \le a < 1$, the powers $a^{n+1}$ tend to zero. Thus

$$\begin{aligned} \sum_{k=0}^{\infty} a^k &= \lim_{n\to\infty} \sum_{k=0}^{n} a^k \\ &= \lim_{n\to\infty} \frac{a^{n+1}-1}{a-1} = \frac{0-1}{a-1} = \boxed{\frac{1}{1-a}}. \end{aligned}$$

For $a > 1$, the partial sums diverge to infinity. When $a = 1$, the series is

$$1 + 1 + 1 + \cdots = \infty.$$

$$\sum_{k=0}^{\infty} a^k = \begin{cases} \dfrac{1}{1-a}, & 0 \le a < 1, \\ \infty, & a \ge 1. \end{cases}$$

**Return to Example 3.2.1: Total Probability**

Apply the geometric-series formula with $a = 1/2$:

$$\begin{aligned}\sum_{x=1}^{\infty} p(x) &= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \sum_{k=1}^{\infty} \left(\frac{1}{2}\right)^k \\ &= \sum_{k=0}^{\infty} \left(\frac{1}{2}\right)^k - \left(\frac{1}{2}\right)^0 \\ &= \frac{1}{1-\frac{1}{2}} - 1 = 2 - 1 = \boxed{1}.\end{aligned}$$

Therefore, $p(x) = (1/2)^x$, $x = 1, 2, 3, \ldots$, is a valid pmf.

**(c) Find the CDF of the Number of Exposed Cells**

$$\text{p.m.f.:} \qquad p(x) = \left(\frac{1}{2}\right)^x, \qquad x = 1, 2, 3, \ldots$$

The CDF is $F(x) = P(X \le x)$. If $x < 1$, no possible value of $X$ is at most $x$, so

$$F(x) = 0 \qquad (x < 1).$$

For example, $F(0) = P(X \le 0) = 0$.

For $x \ge 1$, write $[x]$ for the **greatest integer less than or equal to** $x$. Since $X$ takes only integer values,

$$F(x) = P(X \le x) = P(X \le [x]).$$

For instance,

$$F(2.11) = P(X \le 2.11) = P(X \le 2).$$

Using the Complement Rule,

$$F(x) = 1 - P(X > [x]).$$

**Interpret the complement.** $X$ counts the cells exposed until fusion takes place. Therefore, $X > [x]$ means that the first $[x]$ cells all fail to fuse:

$$\begin{aligned}F(x) &= 1 - P(\text{no fusion in the first } [x] \text{ cells}) \\ &= 1 - P(C_1 \neq F \ \cap \ C_2 \neq F \ \cap \ \cdots \ \cap \ C_{[x]} \neq F) \\ &\overset{\text{SMR}}{=} 1 - \left[P(C_1 \neq F)P(C_2 \neq F)\cdots P(C_{[x]} \neq F)\right] \\ &= 1 - \underbrace{\left[\left(1-\frac{1}{2}\right)\left(1-\frac{1}{2}\right)\cdots\left(1-\frac{1}{2}\right)\right]}_{[x] \text{ times}} \\ &= 1 - \left(\frac{1}{2}\right)^{[x]}.\end{aligned}$$

IND. $\longrightarrow$ SMR

Combining the two cases,

$$\boxed{F(x) = \begin{cases} 0, & x < 1, \\ 1 - \left(\frac{1}{2}\right)^{[x]}, & x \geq 1. \end{cases}}$$

**(d) Find** $P(X \geq 10)$

**Direct sum.** The possible values $10, 11, 12, \ldots$ are mutually exclusive:

$$\text{M.E.} \longrightarrow \text{SAR}$$

$$\begin{aligned} P(X \geq 10) &= P(X = 10 \cup X = 11 \cup X = 12 \cup \cdots) \\ &\overset{\text{SAR}}{=} P(X = 10) + P(X = 11) + P(X = 12) + \cdots \\ &= \left(\frac{1}{2}\right)^{10} + \left(\frac{1}{2}\right)^{11} + \left(\frac{1}{2}\right)^{12} + \cdots . \end{aligned}$$

**OR**

**Using the CDF and Complement Rule.**

$$\begin{aligned} P(X \geq 10) &\overset{\text{CR}}{=} 1 - P(X < 10) \\ &= 1 - P(X \leq 9) \\ &= 1 - F(9) \\ &= 1 - \left[1 - \left(\frac{1}{2}\right)^{[9]}\right] \\ &= \boxed{\left(\frac{1}{2}\right)^{9}}. \end{aligned}$$

**(e) Find** $P(5 \leq X \leq 25)$

**Direct sum.**

$$\begin{aligned} P(5 \leq X \leq 25) &= P(X = 5 \cup X = 6 \cup \cdots \cup X = 25) \\ &\overset{\text{SAR}}{=} P(X = 5) + P(X = 6) + \cdots + P(X = 25) \\ &= \left(\frac{1}{2}\right)^{5} + \left(\frac{1}{2}\right)^{6} + \cdots + \left(\frac{1}{2}\right)^{25}. \end{aligned}$$

**OR**

**Using the CDF.**

$$\begin{aligned} P(5 \le X \le 25) &= P(X \le 25) - P(X < 5) \\ &= P(X \le 25) - P(X \le 4) \\ &= F(25) - F(4) \\ &= \left[1 - \left(\frac{1}{2}\right)^{[25]}\right] - \left[1 - \left(\frac{1}{2}\right)^{[4]}\right] \\ &= \boxed{\left(\frac{1}{2}\right)^{4} - \left(\frac{1}{2}\right)^{25}}. \end{aligned}$$